

Yashvardhan Gupta

+1(408)5910621 ◇ gupta.yashv@northeastern.edu ◇ San Jose, CA, USA ◇ [LinkedIn](#) ◇ [Github](#) ◇ [Website](#)

SUMMARY

AI & Systems Engineer specializing in reinforcement learning, multi-agent architectures, and enterprise-grade pipelines. Experienced in building scalable AI systems, structured data workflows, and production-ready microservices.

TECHNICAL KNOWLEDGE

Languages : Python, Java, SQL, Bash

ML & DL Frameworks : Jax/Flax, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV

Generative & Agentic AI : Transformers, Hugging face, Diffusion models, RAG pipelines, FAISS, LangChain, CrewAI

Systems & Devops : Docker, FastAPI, Flask, CI/CD, GCP, AWS, CUDA, ONNX Runtime, Git

EDUCATION

MS in Artificial Intelligence, Northeastern University (GPA: 4.0/4.0) Sep '25 — Present

- **Relevant Coursework :** Machine Learning, Foundations of Generative AI, Reinforcement Learning

B.Tech in Mechanical Engineering, Delhi Technological University (GPA: 3.3/4.0) Aug '19 — Jul '23

- **Core Courses :** Computer Vision, Machine Learning, Engineering Economics

Certifications - Machine Learning Specialization - Stanford University, Tensorflow Adv. Tech. Specialization- Coursera

PROJECTS

Job Hunter – Autonomous AI (Job Application Agent), Northeastern University [Link](#) Jan '26 — Feb '26

- Developed **Job Hunter**, a persistent autonomous agent that discovers relevant internship postings and programmatically fills application forms using structured field matching with deterministic rules and LLM-fallback.
- Architected a **modular discovery + application pipeline** (scheduler, scraper, scanner, applicator, verifier, reporter) that runs as a macOS daemon, enabling continuous 24/7 operation with automated daily job queueing, prioritization, and reporting and persistent SQLite learning and verification layers.

Jarvis (Personal Autonomous AI Agent), Northeastern University [Link](#) Dec '25 — Feb '26

- Engineered a ReAct-based autonomous agent that interprets natural language prompts to reason about and execute real tasks (terminal commands, browser automation, filesystem operations) with multi-provider LLM integration (Anthropic Claude, OpenAI, NVIDIA NIM).
- Built a modular toolkit with structured safety guardrails, audit logging, and Agent-to-Agent (A2A) communication support, enabling LLM-driven automation workflows while preserving controlled execution and discoverability.

Read my lips (Visual Speech-to-Avatar for Speech Impaired People), Northeastern University [Link](#) Sep '25 — Dec '25

- Built a multimodal pipeline converting silent lip movements → synthesized speech → synchronized avatars.
- Integrated Auto-AVSR, Qwen 0.6B (semantic correction), and FLOAT (flow matching) for end-to-end generation.
- Engineered the system for **Apple Silicon (MPS)** by porting CUDA-centric generative models and optimizing tensor operations, achieving ~200ms VSR inference latency **while maintaining high-fidelity output for real-time applications**.

WORK EXPERIENCE

Khoury College Teaching Assistant - Reinforcement Learning & Sequential Decision Making Jan '26 — Apr '26
Northeastern University San Jose, United States

- Guided graduate students in implementing and debugging core reinforcement learning algorithms (**DQN, PPO, Policy Gradients**), supporting convergence analysis, reward shaping, and training stability in PyTorch-based experiments.
- Conducted recitations and office hours on Markov Decision Processes (MDPs), **exploration–exploitation tradeoffs, and actor-critic methods**, providing structured feedback on experiment design, reproducibility, and **hyperparameter tuning**.

Digital Solutions & Technology Engineer Apr '23 — Apr '25
Biowolk Healthcare Delhi, India

- Engineered a scalable analytics microservice using **Python (FastAPI)** and **Docker** to process daily sales and inventory data, effectively reducing manual reporting time by **15% (~7 hours/week)** via **end-to-end data pipeline orchestration**
- Implemented automated budget-allocation and prediction pipelines (Python), reducing monthly spend variance by 8% through batch retraining and CI/CD model rollout.

PUBLICATIONS

Vision Language Models : A complete survey (Ongoing) Dec '25

Reviews Vision–Language–Action models that combine visual perception, language grounding, and action generation for robotics. The paper identifies strengths, gaps, and opportunities for building next-generation embodied systems.

An End to End Solution to Automated Hiring Dec '22

Proposed and evaluated a GAN-, NLP-, and CV-driven e-recruitment platform automating resume short-listing, deepfake-simulated interviews with dynamic question generation, and CV-based proctoring to accelerate hiring.

<https://ieeexplore.ieee.org/document/10060436>